Department of Software Engineering
Faculty of Science & Information Technology

UniExpert

AI-Powered Expert Discovery and Academic Collaboration Platform

Software Requirements Specification (SRS) — Minimal Edition

Conforming to IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018 Engineering Standard

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Course: SE 331 — Software Engineering Design Capstone Project

Semester: Fall 2026 · **Duration:** October – December 2026

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1 Document Control and Specification Baseline

This minimalist Software Requirements Specification (SRS) establishes the binding technical, functional, and non-functional requirements for **UniExpert** under course **SE 331 (Software Engineering Design Capstone Project)** for Fall 2026.

Ver.	Date	Author(s)	Scope and Summary of Major Changes	Status
0.1	Oct 02, 2026	A. Zafor, A. Haque	Initial requirements elicitation, actor profiles, and scope drafting.	Draft
0.5	Oct 08, 2026	A. Zafor, A. Haque	Complete IEEE 830 functional modules and scoring algorithm definitions.	Reviewed
1.0	Oct 14, 2026	A. Zafor, A. Haque	Minimalist Baseline SRS compiled for Capstone Committee approval.	Approved

Table 1: Document Revision Control

1.1 Executive Overview and IEEE 830 Conventions

UniExpert transforms academic discovery from rigid keyword lookups into an intelligent, problem-first matching workflow. Students describe research goals in natural language; the system extracts semantic skill entities, indexes candidate profiles in vector stores, executes multi-criteria ranking, and manages collaboration through an auditable Finite State Machine (FSM).

- **Requirement Notation:** Functional requirements are identified as `FR-[Module]-[Num]`. Non-functional requirements are identified as `NFR-[Category]-[Num]`.
- **MoSCoW Prioritization:** [**M**] Must-Have (mandatory MVP scope); [**S**] Should-Have (high-priority evaluation feature); [**C**] Could-Have (desirable extension); [**W**] Won't-Have (deferred out-of-scope).
- **Normative Compliance:** RFC 2119 keywords (**SHALL**, **SHOULD**, **MAY**) define requirement binding.

Acronym	Full Term	Operational Definition in UniExpert
EFS	Expert Finding System	Information retrieval system identifying experts based on domain competencies.
HNSW	Hierarchical Navigable Small World	Graph-based vector indexing algorithm in Qdrant for approximate nearest neighbors.
XAI	Explainable Artificial Intelligence	Justification subsystem articulating human-readable match rationales.
FSM	Finite State Machine	Formal state transition model enforcing valid collaboration proposal lifecycles.
RBAC	Role-Based Access Control	Security model segmenting permissions across students, experts, and admins.
NDCG	Norm. Discounted Cumulative Gain	Standard information retrieval metric evaluating graded ranking relevance.

Table 2: Glossary of Operational Acronyms

2 Overall Description and System Context

2.1 System Context Diagram

UniExpert is a decoupled web application interfacing with institutional users, academic bibliometric registries, and vector search engines. Figure 1 illustrates system boundaries and external interfaces.

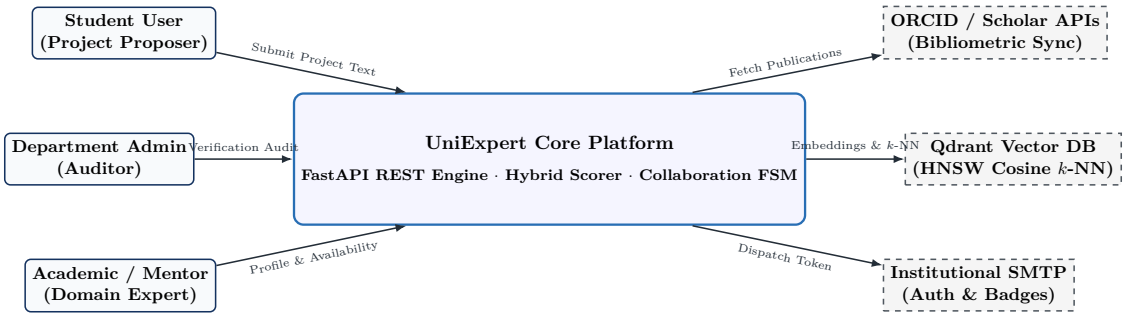


Figure 1: UniExpert System Context and External Interface Model

2.2 User Classes and Operational Privileges

User Role	Authentication Basis	System Capabilities and Operational Permissions	Activity
Student	Verified university email	Submits proposals, inspects matches, forms teams, dispatches collaboration requests.	Daily / Weekly
Expert (Faculty)	Tier 1–3 verified badge	Publishes research skills, configures blackout calendars, accepts/declines requests.	Weekly
Industry Mentor	Tier 4 corporate badge	Reviews applied capstones, provides industry mentoring within set capacity.	Bi-weekly
Administrator	Multi-factor admin account	Approves verification documentation, reviews audit trails, inspects telemetry.	Periodic

Table 3: User Classes and Operational Profiles

2.3 Design Constraints and Operating Environment

- **Hosting & Runtime:** Linux server (Ubuntu 22.04 LTS), Python 3.11+ ASGI (FastAPI), React 18+ Web Client, PostgreSQL 16 relational DB, Qdrant 1.8+ vector DB, Redis 7.2 cache.
- **Inference Optimization:** Zero-shot NLP extraction and dense embeddings (`all-MiniLM-L6-v2`) must run on multi-core CPU architectures via ONNX Runtime without mandatory cloud GPU dependencies.
- **Compliance:** Student proposals and private communications protected by TLS 1.3 in transit and AES-256 at rest, respecting institutional FERPA privacy mandates.

3 Consolidated Functional Requirements Matrix

To maintain high information density, the 20 functional requirements covering all 7 core modules are partitioned into two structured engineering matrices: Table 4 (Core Discovery Modules 1 to 4) and Table 5 (Advanced Collaboration Modules 5 to 7).

Req ID	Feature / Subsystem	Pri.	Normative Functional Specification
Module 1: Authentication and Identity Management			
`FR-AUTH-01`	Registration & Activation	**[M]**	The system SHALL allow students and mentors to register unique accounts. Institutional email domains (`.edu`) SHALL trigger automated HMAC activation tokens (24h TTL).
`FR-AUTH-02`	RBAC & Session Tokens	**[M]**	The system SHALL enforce strict RBAC across `STUDENT`, `EXPERT`, and `ADMIN`. Sessions SHALL use Ed25519-signed JWTs (15m access, 7d HTTP-only refresh).
Module 2: Profile Engineering and Taxonomy Management			
`FR-PROF-03`	Student Academic Profile	**[M]**	The system SHALL maintain student profiles recording department, degree program, research interests, past projects, and CV attachments (PDF, max 5MB).
`FR-PROF-04`	Multi-Tier Credential Badging	**[M]**	The system SHALL enforce a 4-tier verification scheme: Tier 1 (Institutional Email), Tier 2 (ORCID Sync with ≥ 2 papers), Tier 3 (Admin Appointment Audit), Tier 4 (Corporate Domain).
`FR-PROF-05`	Structured Skill Taxonomy	**[M]**	The system SHALL maintain a hierarchical academic skill ontology across engineering, AI, agriculture, and business with automated synonym aliasing (e.g., "CV" $\rightarrow$ "Computer Vision").
Module 3: Natural Language Problem Ingestion and NLP Extraction			
`FR-NLP-06`	Unstructured Problem Ingestion	**[M]**	The system SHALL accept freeform student research statements (50–2,000 words), project title, expected methodology, and target semester timelines.
`FR-NLP-07`	Zero-Shot Entity Extraction	**[M]**	The system SHALL extract domain classifications, technical skill prerequisites, and methodological paradigms using zero-shot classification with confidence scores ≥ 0.60.
`FR-NLP-08`	Interactive Tag Refinement	**[S]**	The system SHALL provide students with an interactive tag manager to adjust weights, prune spurious keywords, and insert custom required competencies prior to search.
Module 4: Semantic Matching and Hybrid Recommendation Engine			
`FR-REC-09`	Vector Embedding Generation	**[M]**	The system SHALL project project texts and expert profiles into 384-dimensional dense vectors using `all-MiniLM-L6-v2` indexed in Qdrant with HNSW graphs ($M = 16$).
`FR-REC-10`	Multi-Criteria Hybrid Scoring	**[M]**	The system SHALL rank experts using composite score $\mathcal{S}(u, e) = \sum_{i=1}^{5} w_i C_i$ (Eq. 1), combining semantic similarity, skill overlap, bibliometrics, tier, and availability.
`FR-REC-11`	Explainable AI (XAI) Justifications	**[M]**	The system SHALL generate human-readable explanations detailing skill intersection counts, relevant publications, and availability rationales for every recommendation.
`FR-REC-12`	Faceted Filtering & Re-Ranking	**[M]**	The system SHALL permit dynamic client-side filtering by verification tier, affiliation, and availability status without requiring re-computation of dense embeddings.

Table 4: Functional Requirements Matrix — Modules 1 to 4 (Authentication, Profiles, NLP, Matching)

Req ID	Feature / Subsystem	Pri.	Normative Functional Specification
Module 5: Multidisciplinary Team Formation and Gap Analysis			
`FR-TEAM-13`	Project Domain Decomposition	**[S]**	The system SHALL automatically decompose interdisciplinary projects into discrete sub-domains (e.g., Computer Vision and Pathology) generating parallel search tracks.
`FR-TEAM-14`	Set-Cover Skill Gap Detection	**[S]**	Given accepted advisory members $\mathcal{M}$, the system SHALL compute unresolved gaps $\Delta\mathcal{K} = \mathcal{K}_u \setminus \bigcup_{e\in\mathcal{M}} \mathcal{K}_e$ (Eq. 2) and recommend complementary recruits.
`FR-TEAM-15`	Side-by-Side Comparison Matrix	**[S]**	The system SHALL render comparative evaluations for up to 3 candidate experts comparing publications, badge tiers, skill overlap percentages, and active advising load.
Module 6: Collaboration Lifecycle and Workspace Management			
`FR-COL-16`	Formal Collaboration FSM	**[M]**	The system SHALL enforce an auditable FSM (`SUBMITTED` → `UNDER_REVIEW` → `ACCEPTED` / `DECLINED` → `ACTIVE` → `COMPLETED`). Illegal transitions SHALL return HTTP 409.
`FR-COL-17`	Anti-Fatigue Quotas & Blackouts	**[M]**	The system SHALL enforce a monthly incoming quota (max 5 active requests per faculty), blackout calendar toggles, and 14-day auto-expiration on unanswered requests.
`FR-COL-18`	Shared Workspace & Messaging	**[M]**	Upon reaching `ACCEPTED`, the system SHALL spin up a shared workspace featuring milestone checklists, file exchange (PDF/ZIP, max 10MB), and audit-logged messaging.
Module 7: Governance, Feedback, and Administration			
`FR-GOV-19`	Double-Blind Peer Review	**[S]**	Upon project completion, both student and mentor SHALL submit mutual 4-axis reviews (responsiveness, mentorship, goal attainment, professionalism) updating profile scores.
`FR-GOV-20`	Administrative Audit Portal	**[M]**	The system SHALL provide departmental admins with a verification queue, document inspector, user dispute resolution console, and system telemetry dashboard.

Table 5: Functional Requirements Matrix — Modules 5 to 7 (Teams, Lifecycle, Governance)

3.1 Mathematical Formulations of Core Engines

(1) Hybrid Recommendation Formulation (FR-REC-10):

$$\mathcal{S}(u,e) = w_1 \cdot \mathcal{D}(u,e) + w_2 \cdot \mathcal{K}(u,e) + w_3 \cdot \mathcal{R}(u,e) + w_4 \cdot \mathcal{V}(e) + w_5 \cdot \mathcal{A}(u,e) \tag{1}$$

where $\mathcal{D}(u,e) = \frac{\vec{v}_u \cdot \vec{v}_e}{\|\vec{v}_u\|\|\vec{v}_e\|}$ is dense semantic cosine similarity, $\mathcal{K}(u,e) = \frac{|\mathcal{K}_u \cap \mathcal{K}_e|}{|\mathcal{K}_u|}$ is weighted skill overlap, $\mathcal{R}(u,e)$ is normalized citation/publication count, $\mathcal{V}(e) \in \{0.25, 0.50, 0.75, 1.0\}$ represents verification tier authority, and $\mathcal{A}(u,e) = 1 - \frac{\text{ActiveLoad}}{\text{MaxLoad}}$ is availability capacity. Baseline weights: $w_1 = 0.30, w_2 = 0.25, w_3 = 0.20, w_4 = 0.15, w_5 = 0.10$ ($\sum w_i = 1.0$).

(2) Advisory Skill Gap Detection (FR-TEAM-14):

$$\Delta\mathcal{K} = \mathcal{K}_u \setminus \bigcup_{e\in\mathcal{M}} \mathcal{K}_e, \qquad \text{where } \mathcal{M} = \{e_1, e_2, \ldots, e_m\} \tag{2}$$

When $|\Delta\mathcal{K}| > 0$, the system executes a greedy set-cover selection to nominate candidate experts $e^* = \arg\max_{e\notin\mathcal{M}} |\mathcal{K}_e \cap \Delta\mathcal{K}|$.

4 Non-Functional Requirements and Industrial Benchmarks

Req ID	Category / Metric	Technical Specification & Acceptance Threshold	Verification
NFR-PERF-01	Search Response Latency	P_{95} semantic retrieval latency SHALL be $\leq$ 250 ms over 10,000 indexed profiles.	k6 Load Test
NFR-PERF-02	NLP Extraction Speed	Entity extraction for a 500-word statement SHALL finish within ≤ 1.2 s on multi-core CPU.	Benchmark
NFR-PERF-03	System Concurrency	API server SHALL sustain 200 concurrent requests with error rate $\leq 0.1\%$.	Stress Test
NFR-PERF-04	Frontend First Render	Web client First Contentful Paint (FCP) SHALL be < 1.0 s over standard 4G broadband.	Lighthouse
NFR-SEC-01	Data in Transit	All communications SHALL be encrypted under TLS 1.3 with strict HSTS enforcement.	SSL Labs (A+)
NFR-SEC-02	Password Security	Passwords salted and hashed with Argon2id ($m = 64\text{MB}, t = 3, p = 4$); plaintext never logged.	Code Audit
NFR-SEC-03	Session Protection	JWTs signed with asymmetric keys; refresh tokens stored in `HttpOnly, SameSite=Strict` cookies.	Security Test
NFR-SEC-04	Rate Limiting	Authentication endpoints throttled to 5 failed attempts per minute per IP via Redis.	Penetration Test
NFR-REL-01	System Availability	Discovery service SHALL guarantee $\geq 99.9\%$ monthly uptime (excluding scheduled windows).	Uptime Monitor
NFR-REL-02	Backup & Recovery	Daily automated PostgreSQL WAL archiving achieving RPO ≤ 1 hour and RTO ≤ 30 minutes.	Disaster Drill
NFR-REL-03	Fallback Degradation	If Qdrant is temporarily offline, system SHALL gracefully degrade to PostgreSQL BM25 keyword search.	Chaos Test
NFR-USE-01	Accessibility Compliance	Web interfaces SHALL comply with WCAG 2.1 Level AA with complete keyboard/screen-reader support.	axe-core Audit
NFR-USE-02	Usability Evaluation	The platform SHALL achieve a System Usability Scale (SUS) score ≥ 80 during Capstone UAT.	User Survey

Table 6: Non-Functional Requirements and Quality Attributes

5 Data Requirements and Relational Data Dictionary

Relational transactional data resides in PostgreSQL 16 under strict foreign key constraints, while 384-dimensional embeddings are indexed in Qdrant collections.

Entity	Attribute Name	Data Type	Constraints and Semantic Purpose
`users`	`id`	UUID	Primary Key, generated via `uuid_generate_v4()`.
	`email`	VARCHAR(255)	Unique, valid institutional format, B-tree indexed.
	`password_hash`	VARCHAR(255)	Argon2id cryptographic hash string.
	`role`	ENUM	`STUDENT`, `EXPERT`, `ADMIN`.
	`is_verified`	BOOLEAN	Default `FALSE`; set `TRUE` upon activation token verification.
`expert_profiles`	`id`	UUID	Primary Key; Foreign Key referencing `users(id)` on delete cascade.
	`institution`	VARCHAR(255)	University or corporate organization name.
	`verification_tier`	INTEGER	Values $1, 2, 3, 4$ representing credential authority badge.
	`availability`	ENUM	`AVAILABLE`, `LIMITED`, `BLACKOUT`.
	`max_load`	INTEGER	Maximum simultaneous active collaborations (default: 3).
`projects`	`id`	UUID	Primary Key.
	`student_id`	UUID	Foreign Key referencing `users(id)`.
	`title`	VARCHAR(255)	Project working title.
	`raw_statement`	TEXT	Natural language problem description (50–2,000 words).
	`embedding_id`	VARCHAR(64)	Vector UUID reference to Qdrant collection point.
`collaborations`	`id`	UUID	Primary Key.
	`project_id`	UUID	Foreign Key referencing `projects(id)`.
	`expert_id`	UUID	Foreign Key referencing `expert_profiles(id)`.
	`state`	ENUM	`SUBMITTED`, `ACCEPTED`, `ACTIVE`, `COMPLETED`, etc.
	`created_at`	TIMESTAMPTZ	Immutable UTC timestamp generated at proposal submission.

Table 7: Relational Database Data Dictionary

6 Requirements Traceability Matrix (RTM)

Table 8 links each functional requirement to its responsible architectural subsystem, test case ID, and standard IEEE verification method: **T** (Test), **D** (Demonstration), **A** (Analysis), and **I** (Inspection).

Req ID	Requirement Title	Architectural Module	Test Case ID	Method
FR-AUTH-01	Registration & Token Activation	Auth Service	TC-AUTH-01	T
FR-AUTH-02	Role-Based Access Control (RBAC)	Gateway Service	TC-AUTH-02	T
FR-PROF-03	Student Profile Management	Profile Service	TC-PROF-01	D
FR-PROF-04	Multi-Tier Credential Badging	Verification Service	TC-PROF-02	I, T
FR-PROF-05	Academic Skill Taxonomy	Taxonomy Service	TC-TAX-01	T
FR-NLP-06	Project Statement Ingestion	Project Service	TC-PROJ-01	D
FR-NLP-07	Zero-Shot Entity Extraction	AI Inference Engine	TC-NLP-01	A, T
FR-NLP-08	Interactive Tag Refinement	Web Client UI	TC-NLP-02	D
FR-REC-09	Dense Vector Indexing	Qdrant Store Service	TC-VEC-01	T
FR-REC-10	Multi-Criteria Hybrid Ranking	Matchmaking Engine	TC-REC-01	A, T
FR-REC-11	Explainable AI Justifications	XAI Service	TC-REC-02	D, T
FR-REC-12	Dynamic Faceted Filtering	Search Service	TC-REC-03	T
FR-TEAM-13	Project Domain Decomposition	Team Service	TC-TEAM-01	A, T
FR-TEAM-14	Set-Cover Skill Gap Analysis	Team Service	TC-TEAM-02	A, T
FR-TEAM-15	Side-by-Side Comparison Matrix	Web Client UI	TC-TEAM-03	D
FR-COL-16	Collaboration Lifecycle FSM	Collaboration Service	TC-COL-01	T
FR-COL-17	Anti-Fatigue Quotas & Blackouts	Governance Service	TC-COL-02	T
FR-COL-18	Shared Workspace & Messaging	Workspace Service	TC-COL-03	D, T
FR-GOV-19	Double-Blind Peer Review	Reputation Service	TC-GOV-01	D
FR-GOV-20	Administrative Audit Dashboard	Admin Service	TC-GOV-02	D, I

Table 8: Requirements Traceability Matrix (RTM)

7 Verification Strategy and Committee Sign-Off

7.1 Testing and Information Retrieval Benchmarking

- **Continuous Integration (GitHub Actions):** Every commit triggers automated Pytest suites verifying mathematical scoring bounds, FSM state validity, and schema migrations with $\geq 85\%$ line coverage.
- **IR Evaluation Protocol:** Recommendation quality is validated against a 50-query academic benchmark with ground-truth expert relevance judgments evaluating Precision@K ($K = 3, 5$) and $NDCG$@5, benchmarked against baseline BM25 lexical search.
- **User Acceptance Testing (UAT):** Scheduled for December 2026 with 20 students and 8 faculty members measuring task completion rate ($\geq 90\%$) and SUS score (≥ 80).

7.2 Academic Committee Review and Approval

The undersigned supervisor and capstone committee members confirm that this Software Requirements Specification (SRS) establishes the agreed baseline for the engineering, evaluation, and grading of **UniExpert** for the Fall 2026 academic semester.

Mr. Md. Shohel Arman
Assistant Professor
Department of Software Engineering
Faculty of Science & Information Technology
Date: _____

Head of Department / Committee Chair
Department of Software Engineering
Faculty of Science & Information Technology
Date: _____